

1.5 功和能 (Work and Energy)

§ 1 功 Work

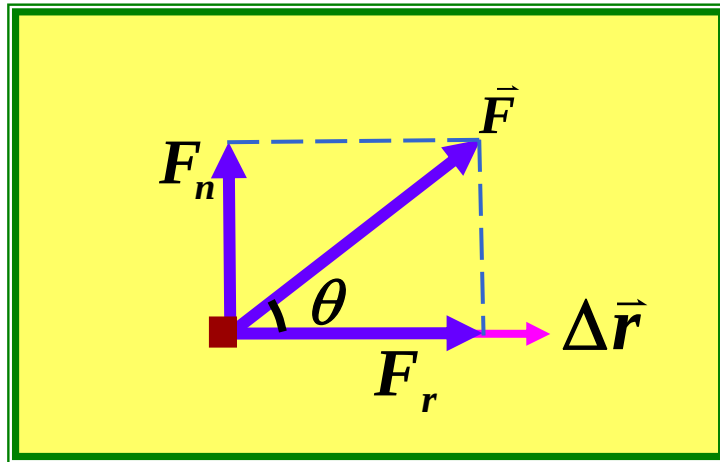
一、功的概念

功的两要素 { 力
在力的方向上的位移

二、恒力的功

$$W = F|\Delta\vec{r}|\cos\theta = F_r|\Delta\vec{r}|$$

$$W = \vec{F} \cdot \Delta\vec{r}$$



讨论: $W = \vec{F} \cdot \Delta\vec{r}$

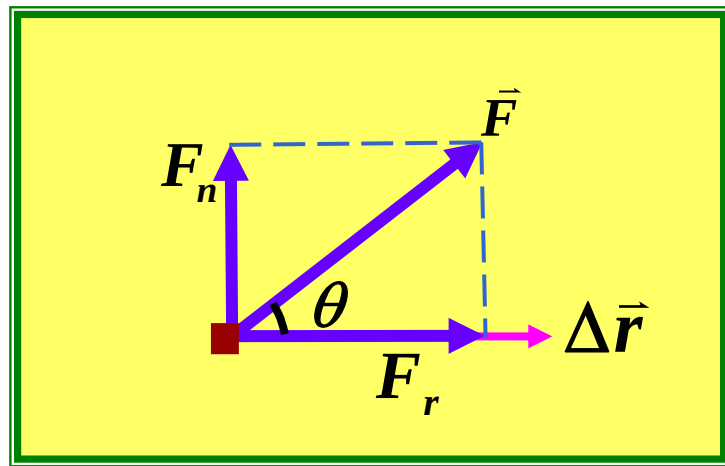
功是**标量**, 但有正负.

单位: 焦耳 $1\text{J}=1\text{N}\cdot\text{m}$

$$0 \leq \theta < \frac{\pi}{2}, W > 0$$

$$\frac{\pi}{2} < \theta \leq \pi, W < 0$$

$$\theta = \frac{\pi}{2}, W = 0$$



直角坐标系中 $W = F_x \cdot \Delta x + F_y \cdot \Delta y + F_z \cdot \Delta z$

三、变力的功

元功 $dW = \vec{F} \cdot d\vec{r}$

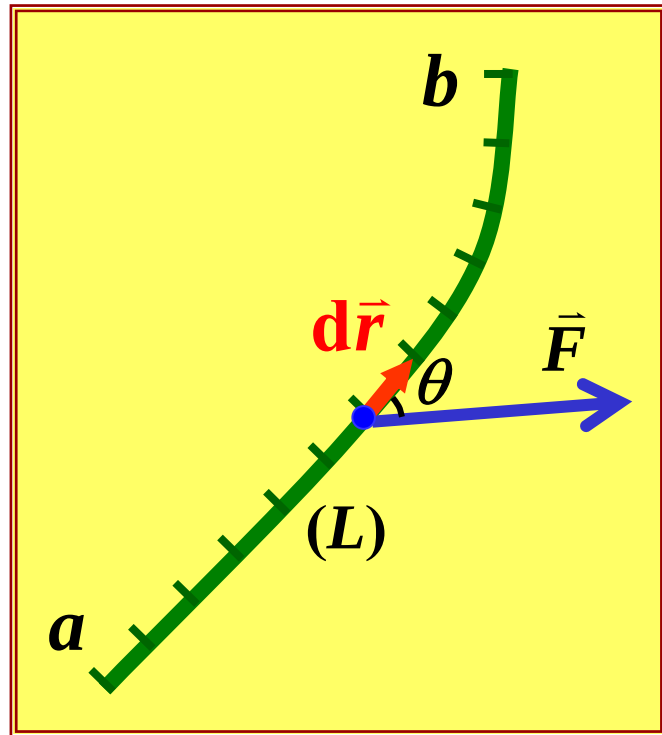
质点从 $a \rightarrow b$ 的功 W

$$W = \int_{a(L)}^b \vec{F} \cdot d\vec{r} = \int_{a(L)}^b F |d\vec{r}| \cos\theta = \int_{a(L)}^b F \cos\theta \, ds$$

直角坐标系中

$$W = \int_{a(L)}^b \vec{F} \cdot d\vec{r} = \int_{a(L)}^b (F_x dx + F_y dy + F_z dz)$$

功的大小与路径相关。



$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

自然坐标系中

$$dW = (F_n \hat{e}_n + F_t \hat{e}_t) \cdot (ds \hat{e}_t) = F_t ds$$

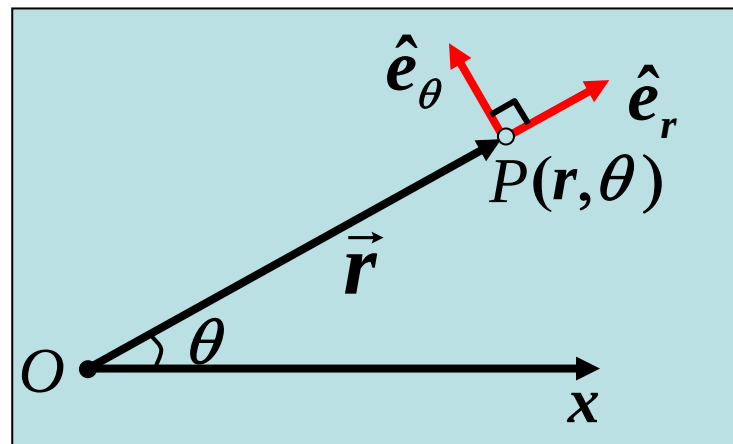
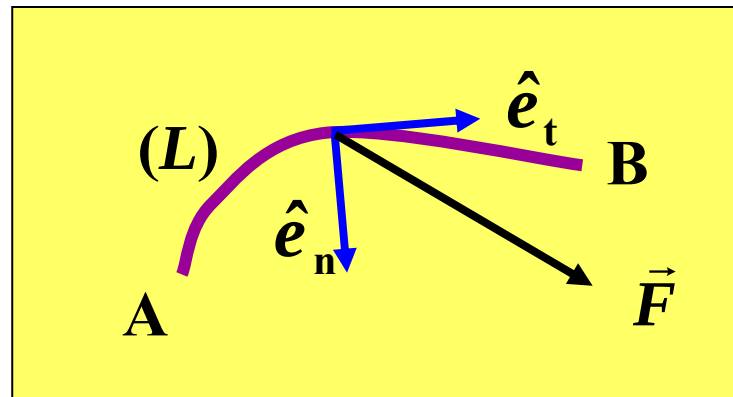
$$W = \int_{A(L)}^B F_t ds$$

极坐标系中

$$dW = (F_r \hat{e}_r + F_\theta \hat{e}_\theta) \cdot (dr \hat{e}_r + r d\theta \hat{e}_\theta)$$

$$W = \int_{A(L)}^B (F_r dr + F_\theta r d\theta)$$

$$dW = \vec{F} \cdot d\vec{r}$$



四、合力的功

质点在几个力的作用下，沿某一路径从 a 运动到 b

$$\vec{F}_1, \vec{F}_2, \dots, \vec{F}_n \quad \text{合力为} \quad \vec{F} = \sum_{i=1}^n \vec{F}_i$$

$$W_{ab} = \int_{a(L)}^b \vec{F} \cdot d\vec{r} = \int_{a(L)}^b (\vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_n) \cdot d\vec{r}$$

$$= \int_{a(L)}^b \vec{F}_1 \cdot d\vec{r} + \int_{a(L)}^b \vec{F}_2 \cdot d\vec{r} + \dots + \int_{a(L)}^b \vec{F}_n \cdot d\vec{r}$$

$$= W_{1ab} + W_{2ab} + \dots + W_{nab}$$

合力的功等于各个力功的和

五、功率 (Power)

单位时间内的功，描述做功的快慢.

1. 平均功率 $\bar{P} = \frac{\Delta W}{\Delta t}$ 单位： 瓦特 (W)

2. 瞬时功率 $P = \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$

$$P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \vec{V} = FV \cos \theta$$

例：质点沿直线运动，质量为 m ，速度为 $v(t) = 3t^2 + 2t \text{ (ms}^{-1}\text{)}$ ，
求： $0 \rightarrow T$ 时间内，外力对此质点做的功。

解： $F = ma = m \frac{dv}{dt} = m \frac{d}{dt}(3t^2 + 2t) = m(6t + 2)$

$$W = \int \vec{F} \cdot d\vec{r} = \int F dx = \int F \frac{dx}{dt} dt = \int F v dt$$

$$= \int_0^T m(6t + 2)(3t^2 + 2t) dt = 2m \int_0^T (9t^3 + 9t^2 + 2t) dt$$

$$= \frac{m}{2} (9T^4 + 12T^3 + 4T^2)$$

例：一物体的质量为 m ，被置于水平桌面上，在外力作用下沿半径为 R 的圆从 A 运动到 B ，移动了半个圆周，设摩擦系数为 μ_k ，求此过程中桌面对它的摩擦力做的功。

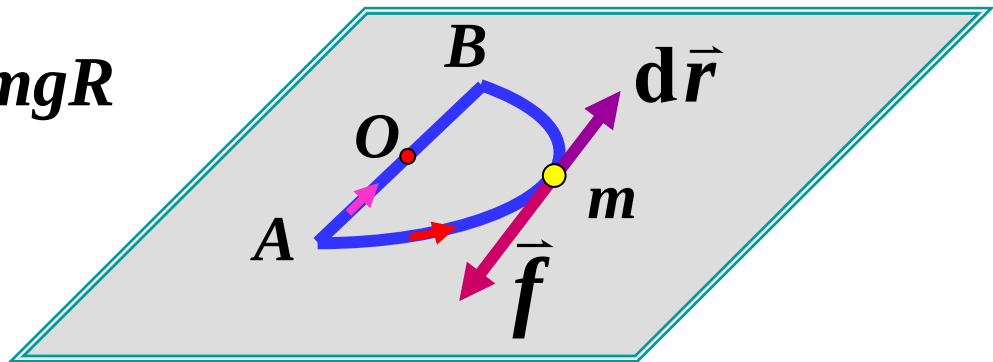
解： $f = \mu_k N = \mu_k mg$

$$dW = \vec{f} \cdot d\vec{r} = -f |d\vec{r}|$$

$$W = \int_A^B -f |d\vec{r}| = \int_A^B -f ds = -\mu_k mg \int_A^B ds = -\mu_k mgs = -\mu_k mg\pi R$$

若沿直径运动： $W = -2\mu_k mgR$

摩擦力的功与路径相关！



例：抛物线 $x^2 = 4y$ 与 直线 $4y = x+6$ 相交于a、b两点，如图所示。作用于某质点上的力为 $\vec{F} = 2y\hat{i} + 4\hat{j}(\text{N})$ 。

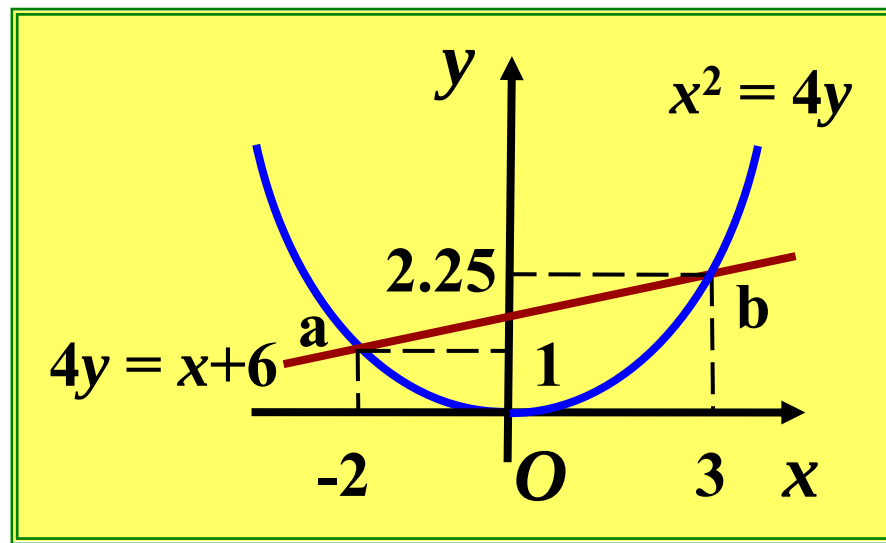
求：质点分别沿抛物线和直线由a 运动到 b 过程中该力的功。

解：抛物线与直线交点的坐标为：a(-2,1), b(3, 9/4)

$$W = \int_a^b \vec{F} \cdot d\vec{r} = \int_a^b (F_x dx + F_y dy)$$

若质点沿抛物线运动

$$\begin{aligned} W_1 &= \int_{x_1}^{x_2} 2y dx + \int_{y_1}^{y_2} 4dy \\ &= \int_{-2}^3 \frac{x^2}{2} dx + \int_1^{9/4} 4dy = 10.8(\text{J}) \end{aligned}$$



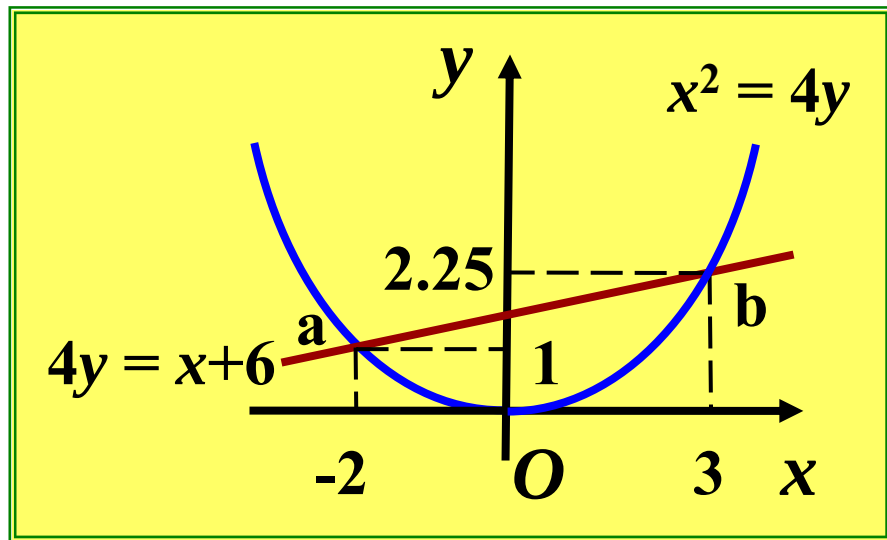
若质点沿直线运动

$$\vec{F} = 2y\hat{i} + 4\hat{j}$$

$$\begin{aligned} W_2 &= \int_a^b (F_x dx + F_y dy) = \int_{x_1}^{x_2} 2y dx + \int_{y_1}^{y_2} 4dy \\ &= \int_{-2}^3 \frac{1}{2}(x+6)dx + \int_1^{9/4} 4dy = 21.25 \text{ (J)} \end{aligned}$$

$$W_1 = 10.8 \text{ (J)}$$

功的大小与运动路径相关



§2 动能定理 Work-energy principle

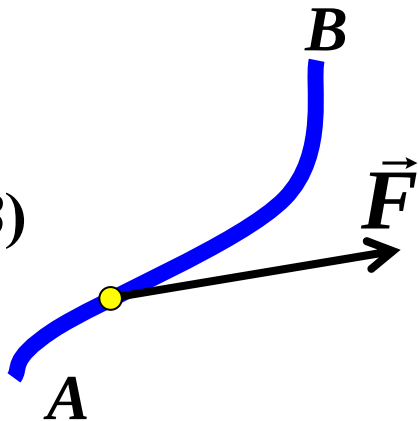
一、质点的动能定理

合外力对质点所做的功等于质点动能的增量

$$W_{AB} = E_{kB} - E_{kA} = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 \quad (A \rightarrow B)$$

$$dW = \vec{F} \cdot d\vec{r} = m \frac{d\vec{v}}{dt} \cdot d\vec{r} = m d\vec{v} \cdot \vec{v} = mv dv$$

$$W_{AB} = \int_A^B dW = \int_{v_A}^{v_B} mv dv = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 = E_{kB} - E_{kA}$$



二、质点系的动能定理

$$m_1 : \int_{A_1}^{B_1} \vec{F}_1 \cdot d\vec{r}_1 + \int_{A_1}^{B_1} \vec{f}_1 \cdot d\vec{r}_1 = \frac{1}{2} m_1 V_{B1}^2 - \frac{1}{2} m_1 V_{A1}^2$$

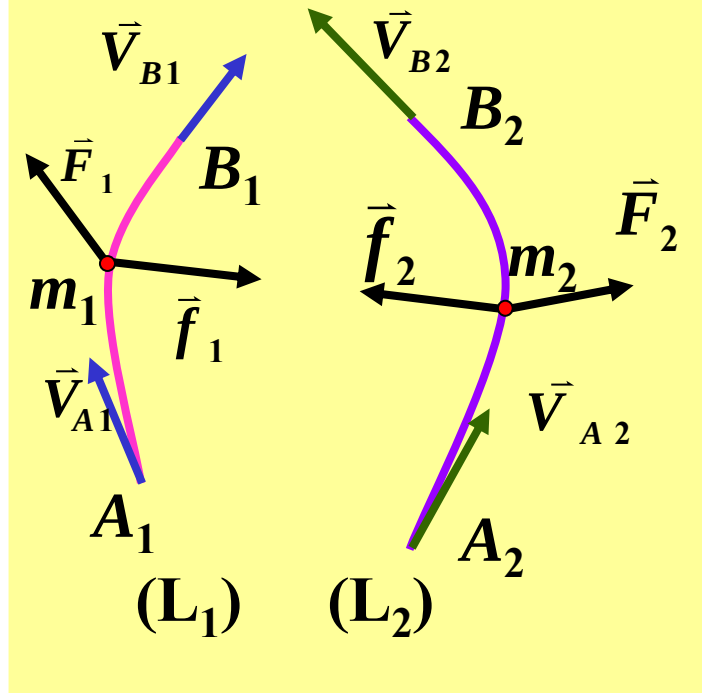
$$m_2 : \int_{A_2}^{B_2} \vec{F}_2 \cdot d\vec{r}_2 + \int_{A_2}^{B_2} \vec{f}_2 \cdot d\vec{r}_2 = \frac{1}{2} m_2 V_{B2}^2 - \frac{1}{2} m_2 V_{A2}^2$$

$$\left(\int_{A_1}^{B_1} \vec{F}_1 \cdot d\vec{r}_1 + \int_{A_2}^{B_2} \vec{F}_2 \cdot d\vec{r}_2 \right) + \left(\int_{A_1}^{B_1} \vec{f}_1 \cdot d\vec{r}_1 + \int_{A_2}^{B_2} \vec{f}_2 \cdot d\vec{r}_2 \right)$$

$W_{\text{外}}$

$W_{\text{内}}$

$$= \left(\frac{1}{2} m_1 V_{B1}^2 + \frac{1}{2} m_2 V_{B2}^2 \right) - \left(\frac{1}{2} m_1 V_{A1}^2 + \frac{1}{2} m_2 V_{A2}^2 \right) = E_{kB} - E_{kA}$$



质点系的动能定理

$$W_{\text{外}} + W_{\text{内}} = E_{kB} - E_{kA}$$

注意

- 内力可以改变质点系的总动能，
尽管内力不能改变质点系的总动量。
- 适用于惯性系。

例：爆竹爆炸过程，内力和为零，
但内力所做的功转化为了动能。

例：同号电荷接近时，动能减少。



三、克尼希定理

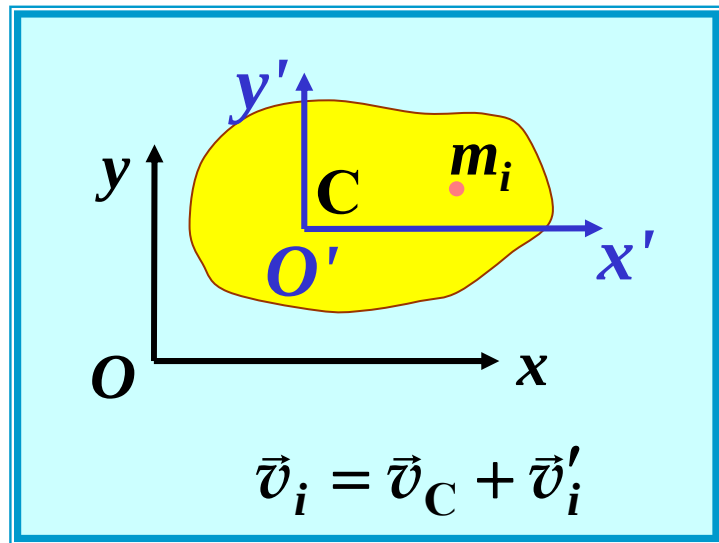
质点系相对于惯性系的动能

$$\begin{aligned} E_k &= \sum \frac{1}{2} m_i v_i^2 \\ &= \sum \frac{1}{2} m_i (\vec{v}'_i + \vec{v}_C) \cdot (\vec{v}'_i + \vec{v}_C) \end{aligned}$$

$$= \sum \frac{1}{2} m_i v_i'^2 + (\sum m_i \vec{v}'_i) \cdot \vec{v}_C + \frac{1}{2} (\sum m_i) v_C^2$$

零

$$E_k = \frac{1}{2} m v_C^2 + \sum \frac{1}{2} m_i v_i'^2 \quad (m = \sum m_i)$$



$$E_k = \frac{1}{2}mv_C^2 + \sum \frac{1}{2}m_i v_i'^2 \quad (m = \sum m_i)$$

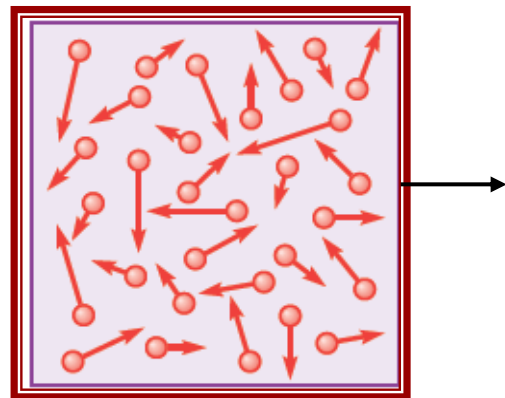
轨道动能: $E_{kC} = \frac{1}{2}mv_C^2$

内动能: 质点系相对于质心系的动能

$$E'_k = \sum \frac{1}{2}m_i v_i'^2$$

克尼希定理: 质点系相对于某一惯性系的动能等于该质点系的轨道动能与内动能之和.

充有气体的箱子



四、 质心系中两质点系统的动能

惯性系 S 中: m_1 的速度为 $\vec{v}_1$, m_2 的速度为 $\vec{v}_2$

$$E_{\text{kC}} = \frac{1}{2}(m_1 + m_2)v_{\text{C}}^2 \qquad \vec{v}_{\text{C}} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

质心系 S' 中: $E'_{\text{k}} = E_{\text{k}} - E_{\text{kC}}$

$$E'_{\text{k}} = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \frac{1}{2}(m_1 + m_2)\left(\frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}\right)^2$$

$$\begin{aligned}
 E'_k &= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \frac{1}{2}(m_1 + m_2)\left(\frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}\right)^2 \\
 &= \frac{1}{2} \cdot \frac{m_1m_2}{m_1 + m_2} (\vec{v}_2 - \vec{v}_1)^2 \quad \text{令2对1的相对速度} \quad \vec{u} = \vec{v}_2 - \vec{v}_1
 \end{aligned}$$

定义： 约化质量(reduced mass) $\mu = \frac{m_1m_2}{m_1 + m_2}$

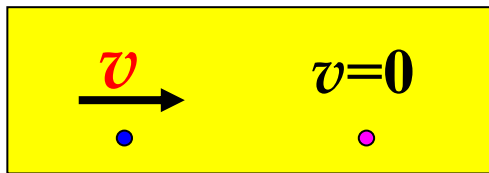
质心系S'中	惯性系S中
$E'_k = \frac{1}{2}\mu u^2$	$E_k = \frac{1}{2}mv_c^2 + \frac{1}{2}\mu u^2$

对于两个粒子 $m_1=m_2=m$ 的碰撞 $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m}{2}$

☺ 其中一个粒子静止

$$E_k = \frac{1}{2} m v^2 \quad u = v$$

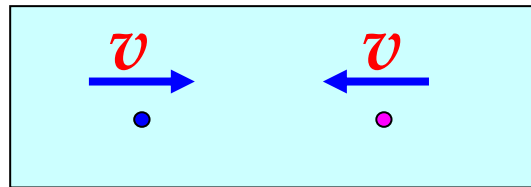
$$\frac{1}{2} \mu u^2 = \frac{1}{4} m v^2 = \frac{1}{2} E_k$$



☺ 两个粒子对撞

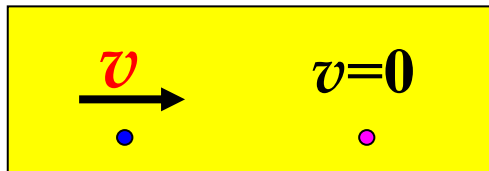
$$E_k = m v^2 \quad u = 2v$$

$$\frac{1}{2} \mu u^2 = m v^2 = E_k$$

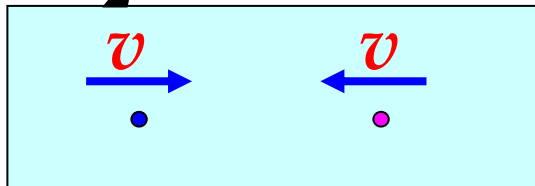


$$E'_k = \sum \frac{1}{2} m_i v_i'^2$$

$$\frac{1}{2} \mu u^2 = \frac{1}{2} E_k$$



$$\frac{1}{2} \mu u^2 = E_k$$



对撞机

核子重如牛 对撞生新态